

Program : Diploma in Tool & Die Engineering	
Course Code : 6119	Course Title: Advanced Tooling Lab
Semester : 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide basic knowledge about parameter settings of injection molding machine & presses.
- To introduce the industrial type assembly & maintenance of various kinds of advanced tooling.

Course Prerequisites:

Topic	Course code	Course name	Semester
Design concept of press tools		Press tool technology	4
Design concept of Molds & Dies technology		Moulds & Dies technology	3
Awareness of advanced Moulding Techniques		Advanced Mould Making	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the working of modern injection moulding machines and presses	9	Applying
CO2	Demonstrate the maintenance & assembly of advanced injection moulds.	12	Applying
CO3	Demonstrate the maintenance & assembly of industrial type press tools	12	Applying
CO4	Demonstrate the maintenance & assembly of industrial type die casting dies	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						3
CO2	3	3					3
CO3	3	3					3
CO4	3	3					3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate modern injection molding machine and presses		
M1.01	Apply the general press settings, die alignment, clamping procedures & shut height adjustments.	3	Applying
M1.02	Operate double action & triple action press.		Applying
M1.03	Apply the Injection molding machine calculations.	3	Applying
M1.04	Discover defective moldings, their causes and remedies.		Applying
M1.05	Operate the general setting like clamping force, cooling time, mold temperature, injection pressure, and holding pressure, knock out rod movement etc. of injection molding machine.	3	Applying
CO2	Demonstrate the maintenance & assembly of advanced injection moulds.		
M2.01	Demonstrate dismantling, Trouble shooting, Maintenance & Assembly of Industrial type split cavity, split core (Angular acting) injection molds.	6	Applying
M2.02	Demonstrate Dismantling, Trouble shooting, Maintenance & Assembly of Industrial type three plate molds, threaded component molds & Hot runner molds.	6	Applying
	Lab Exam – I	1.5	

CO3	Demonstrate the maintenance & assembly of industrial type press tools		
M3.01	Demonstrate dismantling, Trouble shooting, Maintenance & Assembly of Industrial type multi stage progressive tools.	6	Applying
M3.02	Demonstrate dismantling, Trouble shooting, Maintenance & Assembly of Industrial type combination & compound tools.	6	Applying
CO4	Demonstrate the maintenance & assembly of industrial type die casting dies		
M4.01	Discover the causes and remedies of defects – Shrink holes, Gas holes, Segregation, shrink cracks, Porosity, Cold shuts, Flow lines & Blooms, Foliations, Hard spots, Surface draws and Depressions, Soldering, sink mark and excessive flash etc.	4	Applying
M4.02	Demonstrate dismantling, Trouble shooting, Maintenance & Assembly of Industrial type diecasting dies.	5	Applying
	Lab Exam – II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	Pye.R.G.W., "Injection Mould Design", Affiliated East – west press pvt Ltd, 2000
R2	Briston and Gosselin, —Introduction to Plastics", Newnes-Butterworths, London, 1970
R3	Mills.N.J., —Plastics", ELBS, 1986
R4	Dominick V. Rosato and Donald V. Rosato., "Injection Moulding Hand Book", CBS Publishers & Distributors, Delhi, 1987 4
R5	Athalye.A.S., "Moulding of Plastics", Multi Tech Publishing Co., 1998
R6	Donaldson, Tool Design, Tata McGraw-Hill Book company, 23rd edition, 2006
R7	J.R. Paquin, Die design fundamentals, Industrial Press Inc, 1990.

Online Resources:

Sl.No	Website Link
1	https://www.thefabricator.com/stampingjournal/article/stamping/establishing-a-die-setup-recipe-for-progressive-dies
2	https://www.misumi-techcentral.com/tt/en/press/2012/01/109.html
3	https://www.m-ep.co.jp/en/pdf/product/novaduran/molding.pdf
4	https://www.m-ep.co.jp/en/pdf/product/iupi_nova/molding.pdf